

GLOBAL PRODUCT INTERSECTION SETS IN SEMIGROUPS

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ABSTRACT. For a family $(A_q)_{q \in Q}$ of subsets of a semigroup, the product intersection set records those exponents $h \in \mathbb{N}$ for which the h -fold product set of the intersection, $(\bigcap_q A_q)^h$, is equal to $\bigcap_q A_q^h$, the intersection of the product sets. Nathanson recently asked which subsets of $\mathbb{N}$ can occur as a product intersection set, both for arbitrary and for decreasing families $(A_q)_{q \in Q}$. We solve both problems by giving a complete classification. In particular, when $|Q| \geq 2$, we show that in either case any subset $X \subseteq \mathbb{N}$ with $1 \in X$ occurs as a product intersection set. Both classifications were autonomously discovered and formally verified in Lean by [Aristotle](#), a formal reasoning agent developed by [Harmonic](#).

1. INTRODUCTION

Let S be a multiplicative semigroup and, with Q an index set, let $(A_q)_{q \in Q}$ be a family of subsets of S . We then write A for the intersection $\bigcap_{q \in Q} A_q$, where we interpret this intersection to be the entire semigroup if Q is the empty set. Further, for a subset $B \subseteq S$, define the h -fold product set

$$B^h = \{b_1 \cdots b_h : b_1, \dots, b_h \in B\}, \quad h \in \mathbb{N},$$

where throughout the paper, we write $\mathbb{N} := \{1, 2, 3, \dots\}$. With these definitions, one can quickly verify that $A^h \subseteq \bigcap_{q \in Q} A_q^h$ for every $h \in \mathbb{N}$ [2, Eq. (1)]. The set of exponents for which this containment is an equality is called the *product intersection set*

$$H_Q^S(A_q) := \left\{ h \in \mathbb{N} : A^h = \bigcap_{q \in Q} A_q^h \right\}.$$

For a fixed index set Q , the associated global set is

$$\mathbf{H}_Q := \{H_Q^S(A_q) : S \text{ is a semigroup and } (A_q)_{q \in Q} \text{ is a } Q\text{-indexed family of subsets of } S\}.$$

Thus, $\mathbf{H}_Q$ is the collection of all subsets of $\mathbb{N}$ that arise as product intersection sets for Q -indexed families in arbitrary semigroups.

In the special case $Q = \mathbb{N}$ we say that a sequence $(A_q)_{q \geq 1}$ of subsets of a semigroup S is *asymptotically strictly decreasing* if $A_{q+1} \subseteq A_q$ for all $q \geq 1$ and $A_q \neq A_{q+1}$ for infinitely many values of q , i.e. the sequence is not eventually constant. The corresponding global set is

$$\mathbf{H}_{\mathbb{N}}^* := \{H_{\mathbb{N}}^S(A_q) : S \text{ is a semigroup and } (A_q)_{q \geq 1} \text{ is asymptotically strictly decreasing}\}.$$

Nathanson [2, Theorem 14] proved that these two global sets are closed under intersections and asked various natural follow-up questions [2, Problems 10 and 11, and the second part of Problems 4 and 7]:

- Are $\mathbf{H}_Q$ and $\mathbf{H}_{\mathbb{N}}^*$ also closed under unions?
- For a fixed index set Q , does there exist a set $X \subseteq \mathbb{N}$ with $1 \in X$ and such that either $X \notin \mathbf{H}_Q$ or $X \notin \mathbf{H}_{\mathbb{N}}^*$?

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The purpose of this paper is to answer both questions, by exactly describing $\mathbf{H}_Q$ and $\mathbf{H}_{\mathbb{N}}^*$. More precisely, in Section 3 we prove the following result for $\mathbf{H}_{\mathbb{N}}^*$.

Theorem 1.1. *We have*

$$\mathbf{H}_{\mathbb{N}}^* = \{X \subseteq \mathbb{N} : 1 \in X\}.$$

As for $\mathbf{H}_Q$, in Section 4 we show that it is fully determined by the cardinality of Q , in the following way.

Theorem 1.2. *We have*

$$\mathbf{H}_Q = \begin{cases} \{\{1\}, \mathbb{N}\}, & Q = \emptyset, \\ \{\mathbb{N}\}, & |Q| = 1, \\ \{X \subseteq \mathbb{N} : 1 \in X\}, & |Q| \geq 2. \end{cases}$$

We note that answers to Nathanson's two questions immediately follow from Theorems 1.1 and 1.2. Indeed, for both sets the answer to the first question is positive, while the answer to the second question is negative, unless $|Q| \leq 1$ in the case of $\mathbf{H}_Q$. We further remark that for non-empty Q the same results hold true if we restrict to monoids instead of semigroups. To see this, let S be a semigroup and let $(A_q)_{q \in Q}$ be a family of subsets of S , where $Q \neq \emptyset$. Now, if S is a subsemigroup of T , the sets A^h and A_q^h stay the same, regardless of whether they are computed in S or in T . We therefore have the equality

$$H_Q^S(A_q) = H_Q^T(A_q).$$

In particular, if T is the monoid obtained from S by adjoining an identity element, we find that any product intersection set realized by a semigroup can be realized by a monoid as well.

For a brief overview of the proofs, assume for the moment $|Q| \geq 2$ (as the proof for $|Q| \leq 1$ will be fairly straightforward), and let $n \geq 2$ be an integer. Then, in order to prove either Theorem 1.1 or 1.2, we construct an explicit semigroup S_n and an explicit family of subsets $(A_{n,q})_{q \in Q}$ of S_n (which is decreasing in the case of $\mathbf{H}_{\mathbb{N}}^*$) such that

$$H_Q^{S_n}(A_{n,q}) = \mathbb{N} \setminus \{n\}. \quad (1.1)$$

We then combine these single-exclusion building blocks via product constructions to produce arbitrary subsets of $\mathbb{N}$ containing 1. Before we do this, however, we first need a few preliminary lemmas.

2. ELEMENTARY PRODUCT LEMMAS

When $h = 1$, both sides of the containment $A^h \subseteq \bigcap_q A_q^h$ simplify to $\bigcap_q A_q$, so that, in particular, the sets are equal. We therefore obtain the following necessary condition for sets in $\mathbf{H}_Q$ and $\mathbf{H}_{\mathbb{N}}^*$.

Lemma 2.1. *For every set Q and for every Q -indexed family $(A_q)_{q \in Q}$ of subsets of a semigroup S , we have $1 \in H_Q^S(A_q)$. Hence, if $X \in \mathbf{H}_Q$ or $X \in \mathbf{H}_{\mathbb{N}}^*$, then $1 \in X$.*

We now record two basic lemmas for products of semigroups.

Lemma 2.2. *Let $(S_i)_{i \in I}$ be a family of semigroups. For subsets $B_i, C_i \subseteq S_i$, consider the box products*

$$B = \prod_{i \in I} B_i, \quad C = \prod_{i \in I} C_i$$

as subsets of $\prod_{i \in I} S_i$. Then

$$BC = \prod_{i \in I} (B_i C_i).$$

Proof. If $x = bc \in BC$ with $b = (b_i) \in B$ and $c = (c_i) \in C$, then $x_i = b_i c_i \in B_i C_i$ for every i , hence $x \in \prod_i (B_i C_i)$. This proves $BC \subseteq \prod_i (B_i C_i)$.

Conversely, let $x = (x_i) \in \prod_i (B_i C_i)$. For each i choose $b_i \in B_i$ and $c_i \in C_i$ with $x_i = b_i c_i$. Then $b = (b_i) \in B$, $c = (c_i) \in C$, and $x = bc \in BC$. Hence $\prod_i (B_i C_i) \subseteq BC$. $\square$

Corollary 2.3. *With the notation of Lemma 2.2, for every $h \in \mathbb{N}$,*

$$\left(\prod_{i \in I} B_i \right)^h = \prod_{i \in I} B_i^h.$$

Proof. This follows by induction on h . The case $h = 1$ is trivial. If the statement holds for h , then by Lemma 2.2

$$\left(\prod_i B_i \right)^{h+1} = \left(\prod_i B_i \right)^h \left(\prod_i B_i \right) = \left(\prod_i B_i^h \right) \left(\prod_i B_i \right) = \prod_i (B_i^h B_i) = \prod_i B_i^{h+1}. \quad \square$$

Lemma 2.4. *Let I and Q be index sets and let $(A_{i,q})_{i \in I, q \in Q}$ be a doubly indexed family of subsets of semigroups $(S_i)_{i \in I}$. Then*

$$\bigcap_{q \in Q} \left(\prod_{i \in I} A_{i,q} \right) = \prod_{i \in I} \left(\bigcap_{q \in Q} A_{i,q} \right).$$

Proof. An element $a = (a_i)$ belongs to the left-hand side if and only if, for every $q \in Q$ and every $i \in I$, one has $a_i \in A_{i,q}$. This is equivalent to saying that, for every $i \in I$, one has $a_i \in \bigcap_{q \in Q} A_{i,q}$, which is exactly the condition for a to belong to the product on the right-hand side. $\square$

3. THE SET $\mathbf{H}_{\mathbb{N}}^*$

The goal of this section is to prove Theorem 1.1, by first constructing a semigroup S_n and a decreasing sequence of subsets $(A_{n,q})_{q \geq 1}$ of S_n such that (1.1) holds. For this, we define $\mathcal{W}$ to be the set of all non-empty words over the alphabet $\mathbb{N}$, and let α and β be two distinct symbols not contained in $\mathcal{W}$.

3.1. A single-exclusion semigroup. Fix an integer $n \geq 2$, and let $\mathcal{W}_{< n}$ be the subset of $\mathcal{W}$ that consists of all words of length $< n$. We then define $S_n := \mathcal{W}_{< n} \cup \{\alpha, \beta\}$, and we will turn S_n into a semigroup.

For words $u, v \in \mathcal{W}_{< n}$, let uv denote concatenation. We define a multiplication on S_n as follows:

$$\alpha \cdot x = x \cdot \alpha = \alpha, \quad \beta \cdot x = x \cdot \beta = \alpha \quad (x \in S_n),$$

and for $u, v \in \mathcal{W}_{< n}$,

$$u \cdot v = \begin{cases} uv, & |u| + |v| < n, \\ \beta, & |u| + |v| = n, \\ \alpha, & |u| + |v| > n. \end{cases}$$

Lemma 3.1. *S_n is a semigroup.*

Proof. As S_n is certainly closed under the multiplication operation we defined, it is sufficient to prove that this multiplication is associative. Now, if at least one of x, y, z is α or β , then both $(x \cdot y) \cdot z$ and $x \cdot (y \cdot z)$ are equal to α , because any product with one factor equal to α or β equals α . It therefore further suffices to consider the case in which x, y, z are elements of $\mathcal{W}_{< n}$, say $x = u$, $y = v$, $z = w$. With $L := |u| + |v| + |w|$, there are then three cases to consider.

If $L < n$, then both $u \cdot v = uv$ and $v \cdot w = vw$ are words of length $< n$, and

$$(u \cdot v) \cdot w = uv \cdot w = uvw,$$

which is again a word of length $L < n$.

If $L = n$, then $(|u| + |v|) + |w| = |u| + (|v| + |w|) = n$, so that

$$(u \cdot v) \cdot w = u \cdot (v \cdot w) = \beta.$$

If $L > n$, we claim that both $(u \cdot v) \cdot w$ and $u \cdot (v \cdot w)$ are equal to α . To prove $(u \cdot v) \cdot w = \alpha$, first assume $|u| + |v| \geq n$. In that case $u \cdot v \in \{\alpha, \beta\}$, and therefore $(u \cdot v) \cdot w = \alpha$. On the other hand, if $|u| + |v| < n$, then $u \cdot v = uv$ with $|uv| + |w| = L > n$, so that $(u \cdot v) \cdot w = \alpha$. As the symmetric argument applies to $u \cdot (v \cdot w)$, we do indeed get

$$(u \cdot v) \cdot w = u \cdot (v \cdot w) = \alpha.$$

We conclude that the multiplication we defined on S_n is associative, and hence we see that S_n is a semigroup. $\square$

For each $q \geq 1$, define

$$A_{n,q} := \{\alpha\} \cup \{[m] : m \geq q\} \subseteq S_n,$$

where $[m]$ denotes the one-letter word whose only letter is m .

Lemma 3.2. *The sequence $(A_{n,q})_{q \geq 1}$ is strictly decreasing and $\bigcap_{q \geq 1} A_{n,q} = \{\alpha\}$.*

Proof. Clearly $A_{n,q+1} \subsetneq A_{n,q}$ because $[q] \in A_{n,q} \setminus A_{n,q+1}$. Equally clearly, $\alpha \in A_{n,q}$ for all q , while $[m] \notin A_{n,m+1}$, so that $\bigcap_{q \geq 1} A_{n,q} = \{\alpha\}$. $\square$

Lemma 3.3. *Let $n \geq 2$ and $h \in \mathbb{N}$.*

- (i) *If $1 \leq h < n$, then $A_{n,q}^h = \{\alpha\} \cup \{[m_1 \cdots m_h] : m_1, \dots, m_h \geq q\}$.*
- (ii) *If $h = n$, then $A_{n,q}^h = \{\alpha, \beta\}$.*
- (iii) *If $h > n$, then $A_{n,q}^h = \{\alpha\}$.*

Proof. For $h < n$, any product of h elements of $A_{n,q}$ is either α (if one factor is α) or the concatenation of h one-letter words, hence a word of length h all of whose letters are at least q . This proves the inclusion “ $\subseteq$ ”. Conversely, every word $[m_1 \cdots m_h]$ with all $m_i \geq q$ is the product $[m_1] \cdots [m_h]$, so the reverse inclusion holds as well.

For $h = n$, the same argument shows that any product of n factors different from α must be the product of n one-letter words, and by definition such a product is equal to β . Thus $A_{n,q}^n \subseteq \{\alpha, \beta\}$. The reverse inclusion is clear: $\alpha \in A_{n,q}^n$, and $\beta = [q]^n \in A_{n,q}^n$.

If $h > n$, any product containing a factor α is equal to α . On the other hand, if all h factors are one-letter words, then the total length is $h > n$, so the product is α by the definition of multiplication. Hence $A_{n,q}^h = \{\alpha\}$. $\square$

Theorem 3.4. *For every $n \geq 2$,*

$$H_{\mathbb{N}}^{S_n}(A_{n,q}) = \mathbb{N} \setminus \{n\}.$$

Proof. By Lemma 3.2, the intersection of the sequence is $A_n := \{\alpha\}$, and therefore $A_n^h = \{\alpha\}$ for every $h \in \mathbb{N}$.

If $1 \leq h < n$, then by Lemma 3.3(i), an element of $A_{n,q}^h \setminus \{\alpha\}$ is a word of length h whose letters are all at least q ; no such word can belong to the intersection $\bigcap_{q \geq 1} A_{n,q}^h$ over all q . Hence

$$\bigcap_{q \geq 1} A_{n,q}^h = \{\alpha\} = A_n^h.$$

If $h = n$, Lemma 3.3(ii) gives

$$\bigcap_{q \geq 1} A_{n,q}^h = \{\alpha, \beta\} \neq \{\alpha\} = A_n^h.$$

Hence $n \notin H_{\mathbb{N}}^{S_n}(A_{n,q})$.

Finally, if $h > n$, Lemma 3.3(iii) gives

$$\bigcap_{q \geq 1} A_{n,q}^h = \{\alpha\} = A_n^h.$$

Thus $H_{\mathbb{N}}^{S_n}(A_{n,q}) = \mathbb{N} \setminus \{n\}$. □

3.2. Classification of $\mathbf{H}_{\mathbb{N}}^*$. We can now prove our classification theorem on $\mathbf{H}_{\mathbb{N}}^*$.

Proof of Theorem 1.1. By Lemma 2.1, every member of $\mathbf{H}_{\mathbb{N}}^*$ contains 1, so it suffices to prove the converse. Hence, with X a set of positive integers with $1 \in X$, we aim to construct a semigroup S and a strictly decreasing sequence (A_q) such that $H_{\mathbb{N}}^S(A_q) = X$.

If $X = \mathbb{N}$, then we can take S to be the multiplicative semigroup $(\mathbb{N}_0, \cdot)$, with

$$A_q := \{0\} \cup \{m \in \mathbb{N}_0 : m \geq q\} \quad (q \geq 1).$$

Then $(A_q)_{q \geq 1}$ is strictly decreasing, as $q \in A_q \setminus A_{q+1}$. Furthermore, for all $h, q \in \mathbb{N}$ we have both $q \notin A_{q+1}$ and $q \notin A_{q+1}^h$. Hence, for every h we have

$$\bigcap_{q \geq 1} A_q^h = \{0\} = \{0\}^h = \left(\bigcap_{q \geq 1} A_q \right)^h,$$

implying $H_{\mathbb{N}}^{\mathbb{N}_0}(A_q) = \mathbb{N}$, as desired.

Assume now that $X \neq \mathbb{N}$, and let $I := \mathbb{N} \setminus X \subseteq \{2, 3, 4, \dots\}$. For every $n \in I$, let S_n and $(A_{n,q})_{q \geq 1}$ be the building blocks constructed above. We then define the product semigroup

$$S := \prod_{n \in I} S_n$$

and the sequence of subsets

$$A_q := \prod_{n \in I} A_{n,q} \subseteq S \quad (q \geq 1),$$

which is strictly decreasing because any sequence $(A_{n,q})$ with fixed $n \in I$ is.

Now, by Lemmas 2.4 and 3.2 we get

$$A = \prod_{n \in I} \left(\bigcap_{q \geq 1} A_{n,q} \right) = \prod_{n \in I} \{\alpha\},$$

so that, by Corollary 2.3,

$$A^h = \prod_{n \in I} \{\alpha\}^h = \prod_{n \in I} \{\alpha\},$$

for all $h \in \mathbb{N}$. On the other hand,

$$\bigcap_{q \geq 1} A_q^h = \bigcap_{q \geq 1} \left(\prod_{n \in I} A_{n,q}^h \right) = \prod_{n \in I} \left(\bigcap_{q \geq 1} A_{n,q}^h \right).$$

This latter product is equal to A^h if, and only if, all intersections $\bigcap_{q \geq 1} A_{n,q}^h$ are equal to $\{\alpha\}$. By the proof of Theorem 3.4, this happens precisely when $h \notin I$. That is, $h \in H_{\mathbb{N}}^S(A_q)$ exactly when $h \in X$, finishing the proof. □

4. THE SET $\mathbf{H}_Q$

This section will be devoted to determining $\mathbf{H}_Q$.

Proof of Theorem 1.2. Starting off with the easiest case, assume that $|Q| = 1$, say $Q = \{q\}$. Then $A = A_q$, so that in particular $A^h = A_q^h$ for all $h \in \mathbb{N}$. We therefore get that the product intersection set is equal to $\mathbb{N}$, regardless of the semigroup S and the sequence $(A_q)_{q \in Q}$.

Moving on to even fewer elements, let Q be empty and recall that for the empty family of subsets of a semigroup S , the intersection is S itself. Hence the associated product intersection set is $\{h \in \mathbb{N} : S^h = S\}$. We claim that this set is always either $\{1\}$ or $\mathbb{N}$.

To see this, if $S^2 = S$, then by induction $S^h = S$ for all $h \in \mathbb{N}$: once $S^m = S$, we have

$$S^{m+1} = S^m S = SS = S^2 = S.$$

Conversely, if $S^2 \neq S$, then for every $h \geq 2$ one has $S^h \subseteq S^2 \neq S$, so $S^h \neq S$. Thus the only possibilities are $\{1\}$ and $\mathbb{N}$, and both possibilities do actually occur. Indeed, any monoid satisfies $S^h = S$ for all h , realizing $\mathbb{N}$. On the other hand, let $S = \{0, 1\}$ with multiplication $xy = 0$ for all $x, y \in S$. Then $S^h = \{0\}$ for every $h \geq 2$, so this semigroup realizes $\{1\}$.

Finally, we consider the case $|Q| \geq 2$, where we let q_1 and q_2 be two distinguished elements of Q . In analogy with the proof of Theorem 1.1, for each $n \geq 2$ we then construct a semigroup together with a family of subsets, for which the product intersection set is exactly $\mathbb{N} \setminus \{n\}$. So fix an integer $n \geq 2$, and equip the set $S_n := \{0, 1, \dots, n^3 + n^2\}$ with the operation

$$x \star y = \min\{x + y, n^3 + n^2\}.$$

Then $(S_n, \star)$ is a semigroup (a commutative monoid with identity 0, in fact), and by straightforward induction we have the following formula for the h -fold product:

Lemma 4.1. *For all $h \in \mathbb{N}$ and all $a_1, \dots, a_h \in S_n$, we have*

$$a_1 \star \dots \star a_h = \min\{a_1 + \dots + a_h, n^3 + n^2\}.$$

In particular, $a^h = \min\{ha, n^3 + n^2\}$ for all $a \in S_n$.

We now define the subsets $B_n = \{n^2, n^2 + 1\}$ and $C_n = \{n^2, n^2 + n\}$ of S_n , for which we have the following identities for the corresponding powers:

Lemma 4.2. *For all $h \leq n$ we have*

$$B_n^h = \{hn^2 + j : 0 \leq j \leq h\} \quad \text{and} \quad C_n^h = \{hn^2 + jn : 0 \leq j \leq h\},$$

while $B_n^h = C_n^h = \{n^3 + n^2\}$ for all $h > n$.

Proof. If $h \leq n$, then any element of B_n^h is the sum of j copies of $n^2 + 1$ and $h - j$ copies of n^2 , for some j with $0 \leq j \leq h$, which indeed gives

$$B_n^h = \{hn^2 + j : 0 \leq j \leq h\}.$$

On the other hand, if $h > n$, then any sum of h elements of B_n is at least $(n+1)n^2 = n^3 + n^2$, so that B_n^h consists of the single element $n^3 + n^2$. The analogous statements for C_n^h are proved in the same way. $\square$

With the above, the product intersection set is not too hard to determine.

Theorem 4.3. *For every $n \geq 2$,*

$$\{h \in \mathbb{N} : (B_n \cap C_n)^h = B_n^h \cap C_n^h\} = \mathbb{N} \setminus \{n\}.$$

Proof. As $B_n \cap C_n = \{n^2\}$, we have

$$(B_n \cap C_n)^h = \{\min\{hn^2, n^3 + n^2\}\}$$

for every $h \in \mathbb{N}$, by Lemma 4.1. By Lemma 4.2, this agrees with $B_n^h \cap C_n^h$ for $h > n$, but not for $h = n$, where $n^3 + n$ is also contained in $B_n^h \cap C_n^h$. As for $h < n$, we note that any element $hn^2 + jn$ of C_n^h with $j \geq 1$ is larger than any element $hn^2 + j' \in B_n^h$. Hence, for $h < n$ we also have

$$(B_n \cap C_n)^h = B_n^h \cap C_n^h = \{hn^2\}.$$

□

We can now finish the proof of Theorem 1.2. Let X be any set of positive integers with $1 \in X$. For $X = \mathbb{N}$, we can take any monoid S and set $A_q = S$ for all $q \in Q$, which gives $H_Q^S(A_q) = \mathbb{N}$. So assume that $X \neq \mathbb{N}$, and let $I := \mathbb{N} \setminus X \subseteq \{2, 3, 4, \dots\}$. For each $n \in I$, let S_n , B_n , and C_n be as above, and form the product monoid

$$S := \prod_{n \in I} S_n.$$

Further, define subsets of S by

$$B := \prod_{n \in I} B_n, \quad C := \prod_{n \in I} C_n,$$

and let $(A_q)_{q \in Q}$ be defined by

$$A_q := \begin{cases} B, & q = q_1, \\ C, & q = q_2, \\ S, & q \in Q \setminus \{q_1, q_2\}. \end{cases}$$

Then $\bigcap_{q \in Q} A_q = B \cap C$. Moreover, since S is a monoid, $S^h = S$ for every $h \in \mathbb{N}$, and therefore $\bigcap_{q \in Q} A_q^h = B^h \cap C^h$. Thus $H_Q^S(A_q) = \{h \in \mathbb{N} : (B \cap C)^h = B^h \cap C^h\}$.

By Lemma 2.4 and Corollary 2.3,

$$(B \cap C)^h = \left(\prod_{n \in I} (B_n \cap C_n) \right)^h = \prod_{n \in I} (B_n \cap C_n)^h,$$

while

$$B^h \cap C^h = \left(\prod_{n \in I} B_n^h \right) \cap \left(\prod_{n \in I} C_n^h \right) = \prod_{n \in I} (B_n^h \cap C_n^h).$$

Hence

$$h \in H_Q^S(A_q) \iff \forall n \in I, (B_n \cap C_n)^h = B_n^h \cap C_n^h.$$

By Theorem 4.3, the latter is equivalent to $h \neq n$ for every $n \in I$, that is, to $h \in X$. Therefore $H_Q^S(A_q) = X$, and so $X \in \mathbf{H}_Q$ as desired. □

APPENDIX: FORMAL DISCOVERY AND VERIFICATION WITH ARISTOTLE

The proofs in this paper were autonomously discovered and formally verified in Lean by *Aristotle*, a formal reasoning agent developed by [Harmonic](#) [1] and publicly available for free at [aristotle.harmonic.fun](#).

We asked Aristotle to characterise $\mathbf{H}_Q$ and $\mathbf{H}_{\mathbb{N}}^*$ separately, then to combine both characterisations into a complete solution to Problems 10 and 11 in [2]. During the characterisation of $\mathbf{H}_{\mathbb{N}}^*$, our prompt did not specify the meaning of “asymptotically strictly decreasing,” and Aristotle adopted a definition stronger than the one in [2, Section 2]. A simple argument shows that the two definitions yield the same set $\mathbf{H}_{\mathbb{N}}^*$, and the Lean code verifies this equivalence explicitly. We then asked Aristotle to examine whether there were any other problems in [2] that could also be resolved, and it observed that [2, Problems 4 and 7] are in fact consequences of the characterisations of $\mathbf{H}_Q$ and $\mathbf{H}_{\mathbb{N}}^*$ that it had obtained at the outset.

The full Lean formalisation can be inspected interactively in the [Lean 4 Web Editor](#). Aristotle also informalised the Lean file into a first draft of the present paper, which we subsequently rewrote and polished.

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